Roll Number:_____

Thapar Institute of Engineering & Technology, Patiala	
Department of Computer Science &Engineering	
B. E.(COE/CSE) : MST	**Course Code: UCS510/UCS319**
	Course Name: Computer Architecture and Organization
3/Oct/2024	**3:00 AM – 5:00PM**
Time: 2 Hours, M. Marks: 30	**Faculty: SHI/SOL/AAS/YAS/ESH**

Note: All questions carry equal marks. Attempt all questions in order.

<table>
<tr><th>Q.No</th><th>Questions</th><th>Marks</th><th>CO</th><th>BL</th></tr>
<tr><td>Q1.(A)</td><td>Design a logic circuit that controls an elevator door in a three-story building. The circuit has four inputs. M is a logic signal that indicates when the elevator is moving (M = 1) or stopped (M = 0). F1, F2, and F3 are floor indicator signals that are normally LOW, and they go HIGH only when the elevator is positioned at the level of that particular floor. For example, when the elevator is lined up level with the second floor, F2 = 1 and F1 = F3 = 0. The circuit output is the OPEN signal, which is normally LOW and will go HIGH when the elevator door is to be opened. (Design using K- Maps)</td><td>3</td><td>CO1</td><td>L3</td></tr>
<tr><td>Q1.(B)</td><td>Solve showing each step clearly for representing following values $(-102.625)_{10}$ and $(AB.011)_{16}$ in IEEE 754- floating point representation 32 bit precision (convert the final value into hexa-decimal).</td><td>3</td><td>CO1</td><td>L1</td></tr>
<tr><td>Q2.</td><td>Consider the following instruction sequence where registers R1, R2 and R3 are general
purpose and MEMORY[X] denotes the content at the memory location X.

<table>
<tr><th>Instruction</th><th>Operation</th><th>Instruction size (no of words)</th></tr>
<tr><td>MOV R1, (1000)</td><td>R1 <- M[1000]</td><td>1</td></tr>
<tr><td>LOOP: MOV R2, (R3)</td><td>R2 <- M[R3]</td><td>1</td></tr>
<tr><td>ADD R2, R1</td><td>R2 <- R1+R2</td><td>1</td></tr>
<tr><td>MOV (R3), R2</td><td>M[R3] <- R2</td><td>1</td></tr>
<tr><td>INC R3</td><td>R3 <- R3+1</td><td>1</td></tr>
<tr><td>DEC R1</td><td>R1 <- R1-1</td><td>1</td></tr>
<tr><td>if R1 ≠ 0
goto LOOP</td><td>GOTO LOOP on R1 not zero</td><td>1</td></tr>
<tr><td>HALT</td><td>Stop</td><td>1</td></tr>
</table>

Assume that the content of the memory location 1000 is 20, and the content of the register R3 is 2000. The content of each of the memory locations from 2000 to 2020 is 50. The instruction</td><td>3.5</td><td>CO1</td><td>L3</td></tr>
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<tr><td></td><td>sequence starts from the memory location 100. All the numbers are in decimal format. Answer the followings with proper reasoning.

a. Assume that the memory is word addressable. What would be the number of memory references for accessing the data in executing the program completely?
b. Assume that the memory is word addressable. If an interrupt occurs during the execution of the instruction “INC R3” in first iteration, what return address will be pushed on to the stack?
c. Assume the memory is word addressable, after the execution of this program, what would the content of memory location 2020?</td><td></td><td></td><td></td></tr>
<tr><td>Q2.(B)</td><td>Consider a 32- bit processor which supports 67 instructions. Each instruction is 32 bit long and has 5 fields namely mode to specify one of ten addressing mode, opcode, two register identifiers and an immediate operand of signed integer type. Maximum value of the immediate operand that can be supported by the processor is 16383. How many registers the processor has?</td><td>2.5</td><td>CO2</td><td>L4</td></tr>
<tr><td>Q3.(A)</td><td>Design an arithmetic logic unit which can be combined into one ALU with common selection variables such that they can perform following operations for 2 bit ALU.

<table><tr><th>S_2</th><th>S_1</th><th>S_0</th><th>C_{in}</th><th>Operations</th></tr><tr><td>0</td><td>0</td><td>0</td><td>0</td><td>F=A</td></tr><tr><td>0</td><td>0</td><td>1</td><td>1</td><td>F= A+B+1</td></tr><tr><td>1</td><td>1</td><td>0</td><td>X</td><td>F= A $\oplus$ B</td></tr><tr><td>1</td><td>1</td><td>1</td><td>X</td><td>F= A’</td></tr></table></td><td>4</td><td>CO1</td><td>L6</td></tr>
<tr><td>Q3.(B)</td><td>Give 3 logical difference between direct and indirect addressing modes with suitable example.</td><td>2</td><td>CO2</td><td>L1</td></tr>
<tr><td>Q4. a</td><td>A 16-bit basic computer has six input/output (I/O) instructions. The instruction format has three parts, indirect bit (I = 1), decoder operation (D_7 = 1 1 1), and I/O operation. The bit number B_{11} to B_6 in the I/O operation specifies each I/O instruction that is INP, OUT, SKI, SKO, ION, and IOF respectively. The instruction is stored at the memory location 679 (hexadecimal). The initial content of the accumulator (AC) and INPR in hexadecimal is C2F9 and A5. The initial content of FGI is 0 and FGO is 1.
a) Write the executed micro operation performed by each instruction at T_3 clock cycle.</td><td>3</td><td>CO2</td><td>L2</td></tr>
<tr><td>Q4.b</td><td>Determine the content of PC, AR, IR, AC, INPR, OUTR, FGI, FGO IEN, and sequence counter (SC). Consider each I/O instructions independently executed. Give the answer in a table with ten columns, a column representing the value of the register or flag and each row representing the I/O instruction</td><td>3</td><td>CO2</td><td>L3</td></tr>
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Q5.(A)	Assume 6-bit registers that hold the signed numbers. Show the step-by-step multiplication process using the Booth Algorithm where multiplicand is $(-9)_{10}$ and multiplier is $(23)_{10}$.	4	CO3	L5
Q5.(B)	Explain execution micro-operations for BSA and ISZ instructions.	2	CO2	L2



